

ICT702 Data Visualisation

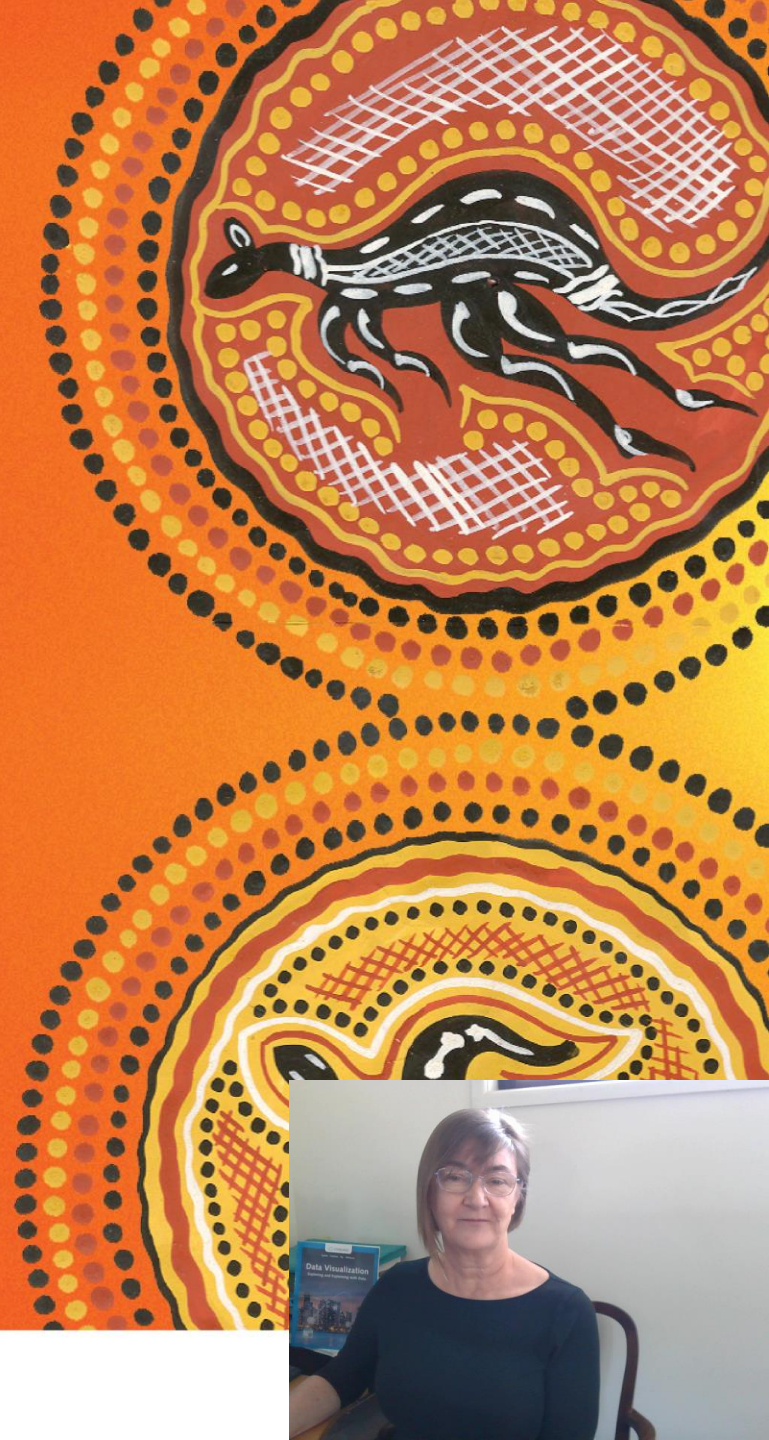
Data Dashboards

Module 8





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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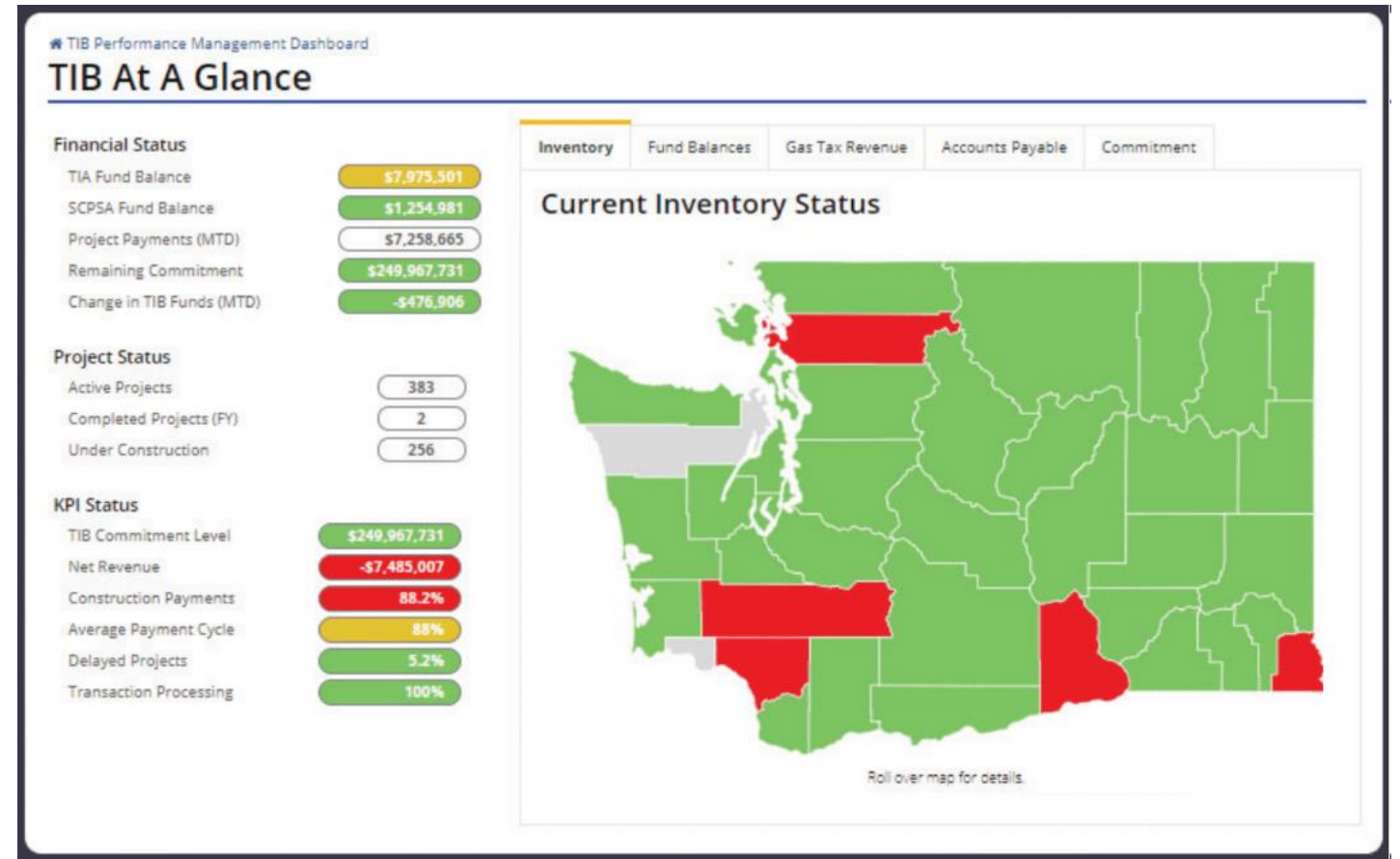
Learning Objectives for Module 8

- Explain what a data dashboard is
- Describe and explain the principles of effective data dashboards
- Understand common areas of application of data dashboards
- Describe and explain various data dashboard taxonomies
- Describe and explain the principles of data dashboard design and development
- Describe and use Excel tools to build a data dashboard
- List common mistakes made in data dashboard design and development

Part 1

Data Visualization Makeover

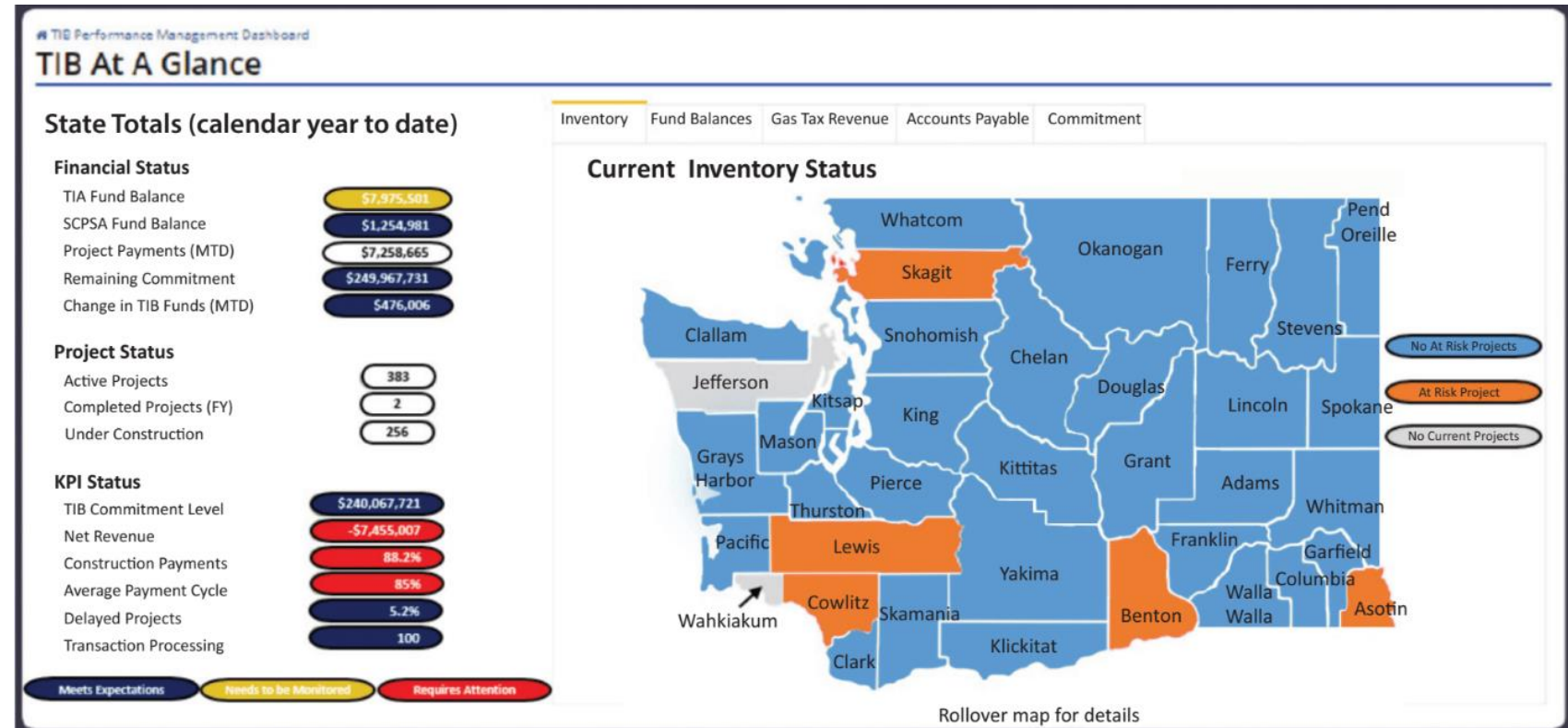
At A Glance Page of the Transportation Improvement Board (TIB) performance management dashboard



A few targeted modifications can make substantial improvements to the dashboard.

Data Visualisation Makeover

At A Glance Page of the improved TIB performance management dashboard



What is a Data Dashboard?

A **data dashboard** is a data visualisation tool that gives multiple outputs and may update in real-time.

Possible **Key performance indicators**:

- the organization's financial position
- inventory on hand
- pending orders for parts
- progress on projects
- status of accounts
- pending orders from customers
- customer service metrics.

Principles of Effective Data Dashboards

- A data dashboard should present all KPIs related to some aspect of the organization's operations on a single screen.
- Create multiple data dashboards of related KPIs rather than having the user scroll vertically or horizontally for information.
- The KPIs displayed in the data dashboard should quickly and clearly convey meaning to its user and be related to decisions the user makes.
- All components of a data dashboard should adhere to the principles of effective data visualisation discussed throughout this book.
- Use colour judiciously to call attention to specific values and to differentiate between categorical variables.

Investment Dashboards

Investment dashboards monitor an organization's performance or individual portfolio of investments over the whole portfolio by individual investment or type of investment. Information provided may include:

- the current price, the recent rate of return, and the relative rate of risk of the portfolio and its components relative to the overall market
- the current price, the recent rate of return, and relative risk of investments under consideration
- the share of the portfolio devoted to various industries or types of investments
- key financial data for the companies in which the investor has invested or is considering investing

Manufacturing Dashboards

Manufacturing dashboards monitor a production process overall and by facility, product, individual, machine, shift, or department. Information provided may include:

- the quantity produced
- the quality of units produced
- the units' production speed
- the frequency of problems (machine downtime, returns, falling short of production targets, etc.)

Marketing Dashboards

Marketing dashboards monitor sales and promotional efforts overall and by product or service, individual, campaign, division or department, or customer. Information provided may include:

- the quantities sold and delivered
- success rate/proportion of successful sales calls
- sales revenue and product and service's selling prices
- advertising and promotional expenditures
- exposure and response to advertising and promotion efforts
- contact with prospective customers
- consumer satisfaction metrics

Human Resource Dashboards

Human resource dashboards monitor the performance of an organization's workforce by individual, division, department, or shift. Information provided may include:

- the number of employees and overall labor effectiveness
- employee churn (rate of employee turnover)
- length of employment
- employee performance and satisfaction
- absenteeism
- participation in training
- reasons for leaving

Technical Support Dashboards

Technical support dashboards monitor the performance of an organization's support of its technology users by the individual, computer system, division or department, or location. Information provided may include:

- type of technical problem
- response time and time to resolution
- outstanding issues
- persistent problems
- downtime
- scheduled maintenance
- software and hardware usage

Personal Fitness/Health Dashboards

Personal fitness/health dashboards monitor aspects of an individual's physical well-being. Information provided may include:

- pulse and blood pressure
- body temperature
- weight and body-mass index (BMI)
- amount of exercise
- number of steps, distance traveled, and calories burned
- diet characteristics (calories, fat intake, carbohydrate and protein intake)
- blood sugar levels

Donor Dashboards

Donor dashboards monitor donor/contributor activity and potential for nonprofit organizations by donor/contributor, prospective donor/contributor, or project/campaign. Information provided may include:

- number and dollar value of donations/contributions
- contact with new potential donors/contributors and past donors/contributors
- expenditures
- conversion rate
- cash flow
- current status of campaigns

Data Dashboards Taxonomies: Data Updates

A **static dashboard** provides information on an organization's KPIs to be manually updated as new data and information become available.

- inexpensive and easy to develop
- helpful when the organization's KPIs change slowly
- updated infrequently and require substantial effort to update

A **dynamic dashboard** provides information on an organization's KPIs and automatically updates as it receives and incorporates new and revised data.

- take time and effort to develop
- useful when the organization's KPIs change rapidly
- updated frequently and require little effort to update

Data Dashboards Taxonomies: User Interaction

A **noninteractive dashboard** does not allow users to customize the data dashboard display. It is useful when the data do not change frequently.

An **interactive dashboard** allows users to customize the data dashboard display using several features:

- **Drill-down** provides more specific and detailed information on a particular element, variable, or KPI.
- **Hierarchical filtering** allows the restriction of the data displayed to specific users by selecting nested values of several categories or values of variables.
- A **time interval widget** allows users to specify the time period to be displayed.
- **Customization tools** allow users to select/deselect/combine variables and categories to be displayed, as well as add/hide custom text and statistics.

Data Dashboards Taxonomies: Organizational Function

Depending on its purposes and ultimate users, data dashboards may belong to one of the following four groups:

- **Operational dashboards** allow lower-level managers to monitor rapidly changing critical business conditions.
- **Strategic dashboards** allow executives to monitor the status of KPIs' relevant overarching organizational objectives.
- **Tactical dashboards** allow mid-level managers to identify and assess the organization's strengths and weaknesses to develop organizational strategies.
- **Analytical dashboards** allow analysts to identify and investigate trends, predict outcomes, and discover insights in large volumes of data.

End of Part 1